TOOL DESIGN

Subject Code : 23A03504a (Professional Elective - IV)

UNIT III : DESIGN OF JIGS AND FIXTURES

Syllabus (as given)

Introduction - Fixed Gages - Gage Tolerances - selection of material for Gauges - Indicating Gages - Automatic gages - Principles of location - Locating methods and devices - Principles of clamping - Drill jigs - General considerations in the design of drill jigs - Drill bushings - Methods of construction - Types of Fixtures - Vice Fixtures - Milling Fixtures - Boring Fixtures - Broaching Fixtures.

1. Introduction

- *Jig:* a work-holding device that locates, holds and also guides the cutting tool (drill bush). It is generally not fixed to the table - e.g. drill jig, reaming jig, tapping jig.
- *Fixture:* a work-holding device that locates and holds the work but does not guide the tool ; it is bolted to the machine table ; the tool is set with a set block / feeler gauge - e.g. milling, turning, grinding, boring, broaching fixtures.

Point	Jig	Fixture
Function	locates, holds and guides tool	locates and holds only
Tool guiding	by drill bushes	set block and feeler gauge
Fixing	not bolted (hand-held or on table)	bolted/clamped to machine table
Used on	drilling, reaming, tapping machines	milling, shaping, planing, grinding, turning
Weight	light	heavy and rigid (absorb cutting forces)

- *Advantages of jigs and fixtures:* interchangeability and accuracy; no marking-out; faster production and lower cost per part; semi-skilled operator; reduced inspection; safe; can use machine to its full capacity.

- Elements: (1) body / base, (2) locating elements, (3) clamping elements, (4) tool guiding (jig) or setting (fixture) elements, (5) fasteners, feet, clamps.

2. Gauges - Fixed Gages

- Gauge: an inspection tool used to check whether a dimension lies within its limits - gives "pass or fail" (go / no-go) information, not the actual size.
- Fixed (limit) gauges: have fixed dimensions corresponding to the limits of size.

2.1 Taylor's principle

- GO gauge should check all dimensions of the feature at the same time (full-form gauge: complete length of engagement, full circular form).
- NO-GO gauge should check only one dimension at a time (point contact) so that a fault in any single dimension is detected.

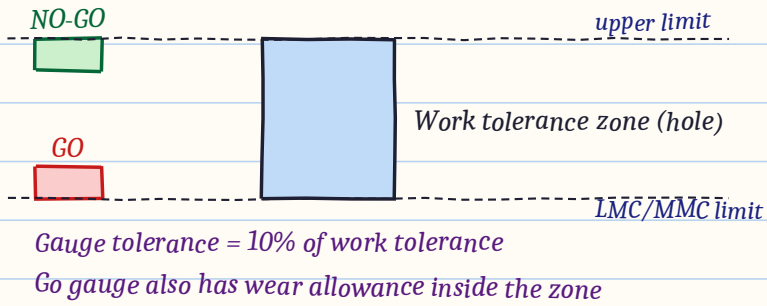
Types of fixed gauges

- Plug gauge (GO-NO GO) - for holes; double-ended or progressive; taper plug gauge
- Ring gauge and snap (gap/caliper) gauge - for shafts; adjustable snap gauge
- Thread plug / ring gauges, taper gauges, spline gauges
- Height, depth, flush-pin gauges , template and profile gauges, feeler (thickness) gauges
- Functional gauges / check fixtures - check position of holes, concentricity

- Working gauge (used by the operator in the shop), inspection gauge (used by inspector, slightly tighter), master (reference) gauge (to check the working gauges - the most accurate).
- Typical design details: GO plug is longer (full length), NO-GO is shorter and marked with a red band; handles (taper lock / trilock type); hardened, ground and lapped.

2.2 Gauge tolerances

- Gauges are themselves made to tolerances, which must be small compared to the work tolerance .



Gauge tolerance = 10% of work tolerance
Go gauge also has wear allowance inside the zone

Gauge tolerance (10% rule)

Work tol. = 0.05 mm
-> gauge tol. = 0.005 mm
Go gauge: tolerance zone placed inside the work limit (MMC side)
No-Go: placed to just stay within the work limit (LMC side)

- *Rule of 10 %:* gauge-maker's tolerance = 10 % of the work tolerance (for work tolerance of IT6 and above); this is split equally between GO and NO-GO gauges.
- *Wear allowance:* GO gauges wear in use; provide a wear allowance (about 5 % of work tolerance, or 0.0025 mm min) *inside the work limit* ; NO-GO gauges do not wear (rarely used) so no wear allowance.

Example

- Hole 25 H7: +0.021 / 0 mm -> work tolerance 0.021 mm
- Gauge tolerance = 10 % of 0.021 = 0.0021 mm; wear allowance = 5 % = 0.0011 mm
- GO plug: 25.0011 to 25.0032 (zone sits inside the work limit by the wear allowance)
- NO-GO plug: 25.0189 to 25.0210 (zone ends exactly on the upper limit)

- Inspection gauges: tolerance 5 % of work tolerance; master gauges: 1/10 of working gauge tolerance (tolerance zone of 10 % of 10 %).

2.3 Selection of material for gauges

- *Requirements:* hardness (>= 62 HRC), wear resistance, dimension stability (no distortion/ageing), good machinability / lapping, corrosion resistance, low thermal expansion, low cost.

Material	Remarks
High carbon high chromium (D2) / oil hardening steel	common, 60-64 HRC; stabilised and lapped
Case hardened MS (EN-8/ 15C8)	large gauges, low cost, case depth 0.8-1.2 mm; wear surface hard, core tough
Chromium plated steel	chrome layer 0.02 mm gives wear resistance and can be re-plated to restore size

Material	Remarks
Tungsten carbide (inserts)	very long life (30-50 times steel) for high volume work
Ceramic / glass / invar	for masters; very stable; low expansion
Aluminium / plastic	only for large light gauges (handles)

- Stabilisation: after hardening, gauges are cold treated (-80 C) and aged / tempered to remove retained austenite and avoid dimension changes.

3. Indicating and Automatic Gages

3.1 Indicating gauges

- Indicating gauge shows the actual deviation of the dimension from the nominal size on a scale/dial - a comparator.
- Common types:

1. Dial indicator (dial gauge): rack and pinion amplification; least count 0.01 mm (0.001 mm for precision).
2. Mechanical comparator (Reed type, Johansson Mikrokator): reed or twisted strip gives high magnification (x 1000-5000).
3. Optical comparator (optimeter): tilting mirror + lens, magnification x 800.
4. Electrical / electronic (LVDT) comparators: LVDT output amplified and shown as digital reading; resolution to 0.1 um .
5. Pneumatic (air) gauges: back-pressure or flow of air from nozzles shows clearance between nozzle and work; non-contact , magnification x 5000-10000; good for bore diameter, taper and out-of-round.

- Uses: checking size, ovality, taper, runout, flatness; fitted on snap, bore and depth gauge bodies; set with slip gauges (gauge blocks) or master pieces.
- Advantages: gives magnitude of error (statistical control possible); one gauge for many sizes; less operator skill. Disadvantages: costlier, slower, needs calibration, delicate.

3.2 Automatic gauges

- Automatic gauging: parts are gauged automatically on the machine or on the line and sorted or the machine corrected without the operator.
- Types:
  - Post-process (sorting) gauges: every part passes through gauge heads which classify it into 'oversize / good / undersize' bins by signals, e.g. ball-bearing sorting.

  - In-process gauges: measure while the part is machined (e.g. grinding): the signal stops the feed when the size is reached.
  - Post-process feedback gauges: gauge controls the offset/tool compensation of the machine (SPC loop).
- Sensors used: electronic LVDT heads, air-electric converters, capacitive and laser/optical sensors, vision systems , touch probes on CNC machines.
- Advantages: 100 % inspection at production speed, no human error, process control and statistics, lower scrap; limitation: costly, needs maintenance and calibration, justified for high volumes.

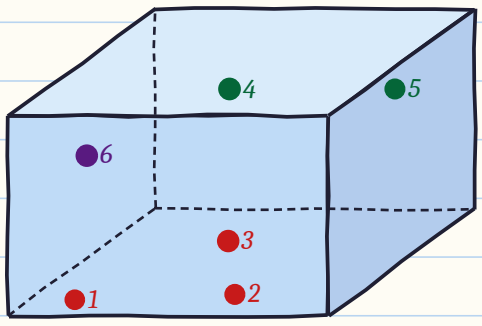
Gauges: quick compare

- Fixed gauge: fast, cheap, only pass/fail
- Indicating gauge: shows deviation, slow but precise
- Automatic gauge: fast + records + controls the machine

4. Principles of Location

4.1 Degrees of freedom

- A free body has 12 degrees of freedom : 6 axial (translation +/- along X, Y, Z -> 6 directions) and 6 radial (rotation clockwise + counter-clockwise about X, Y, Z).
- Locating means restricting some of these so that every workpiece takes the same position in the jig.



3-2-1 (six point) location principle

Points 1,2,3 : base (locating) plane
-> restrain 3 DOF (1 translation + 2 rotations)

Points 4,5 : guiding (back) plane
-> restrain 2 DOF (1 translation + 1 rotation)

Point 6 : stop (side) plane
-> restrain 1 DOF (1 translation)

Total = 6 DOF removed (3 + 3 rotational)

- 3-2-1 principle: locate with 3 points on the base (main) plane, 2 on the second (guide) plane, 1 on the third (stop) plane -> removes 9 of the 12 (the other 3 are removed by clamping force pressing the part against the locators; e.g. the 3 opposite directions).

- Redundant location (using more points than necessary, e.g. 4 points on base) causes the part to rock; use adjustable or equaliser pads where a casting surface is rough.
- Cylindrical part: a long V-block (or 2 short V-blocks) removes 4 DOF (2 translations + 2 rotations); an end stop removes 1 (axial); a key / flat peg stops rotation (1) -> total 6. Circular plate: 3 pads on the face, a central pin (2) and a side peg (1).

4.2 Choosing locating surfaces (datums)

- Select machined surfaces of the part whenever possible; if a casting surface, use 3 fixed points on the largest rough face (spherical head pins) or adjustable (screw jack) supports.
- Select locating surface close to the surface being machined ; the largest spread of the locators gives best stability and accuracy.
- Follow the drawing datum (design datum = locating datum) - this prevents tolerance stack-up and "locating error".
- Locators must be hard, wear resistant, replaceable , and easy to clean (chips can not collect).

5. Locating Methods and Devices

5.1 Locating from flat surfaces

- Fixed rest buttons / pads: flat or spherical headed - spherical buttons for rough cast faces, flat-headed pads for machined faces (dia 8-20 mm). Press-fit H7/p6, hardened 55-60 HRC.
- Adjustable supports (screw jack, spring-loaded): used for uneven castings, tall parts; locked after adjusting.
- Equalising (levelling) supports: a pivoted rocker that touches at more than 2 points; takes unevenness.
- Locating plates / rails; steps; nest (recess) matching outer profile of the part (a nest = a profile pocket, good for thin or small parts, with relief at corners and the depth about 1/2 of the thickness).
- Pins and diamond pins from existing holes (2 holes, one round + one diamond , avoids binding because of centre distance tolerance).

5.2 Locating from cylindrical surfaces (external)

- V-locators (V-block): 90 deg standard (60, 120 also used); centre the cylinder irrespective of diameter variation; the centre line shift vertical = delta d / (2 sin(V/2)); long V gives

4 point contact.

- Cup / cone locators, pad & plug: centralise the shaft end.
- Locating by nest / ring (sleeve), 3-jaw chuck, collet, expanding bush.

5.3 Locating from holes (internal)

- Plug (button) locators: round, with head ; clearance fit H7/g6 (slip) or press fit H7/p6; chamfer of the lead 1.5 mm; for accuracy fixed, hardened.
- Relieved (diamond) pin: the second pin has two flats cut away in the direction of the line joining the pins, leaving a narrow band (b ~ 3-6 mm) across it, so the variation in the centre distance of the two holes is absorbed and the part never binds.
- Expanding mandrels, tapered mandrel (1:2000 to 1:1500), cone locators, conical plug , and pull dowel .

Locating errors

- Error = clearance between locator and hole (max. hole - min. pin)
- Locating pins must be shorter than the hole depth and chamfered
- Hole location of 2 pins: centre distance tolerance of jig = (tolerance of part - clearance)/2
- Use sliding dowels ("pull") to compensate and an ejector for removing

5.4 Locating from profile and other methods

- Pin location from profile: two pins on a straight edge + 1 for the stop (3-2-1).
- Gap / fork locators; screw-type self-centring devices (concentric clamping: scroll, wedge, taper collet).
- Self-centring: all moving; mean position kept regardless of size, error reduced.
- Fool-proofing (poka-yoke): make the location asymmetric (extra pin or block) so the part can be inserted in the right way only.

6. Principles of Clamping

- Clamping is to hold the located workpiece firmly against the locators against cutting forces, weight and vibration without distorting it.

Rules of clamping

1. Clamping force should act on the locators (directed towards the locating surface, never away from them)
2. Clamp should press at rigid portions of the work , above a support , to avoid distortion
3. Force must be just enough to resist tool forces and not exceed; avoid marking finished surfaces (use soft pads)
4. Cutting forces must be taken by the locators and fixed jaws , not by the clamp
5. Should be quick, easy to operate - loading/unloading time is non-productive time
6. Self-releasing is avoided - clamp must not loosen with vibration (self-locking)
7. Clamp must be safe , not sharp, give clearance for the tool , chip and the operator's hands
8. Simple, strong, fool-proof , low cost, using standard parts (hooks, nuts, studs)

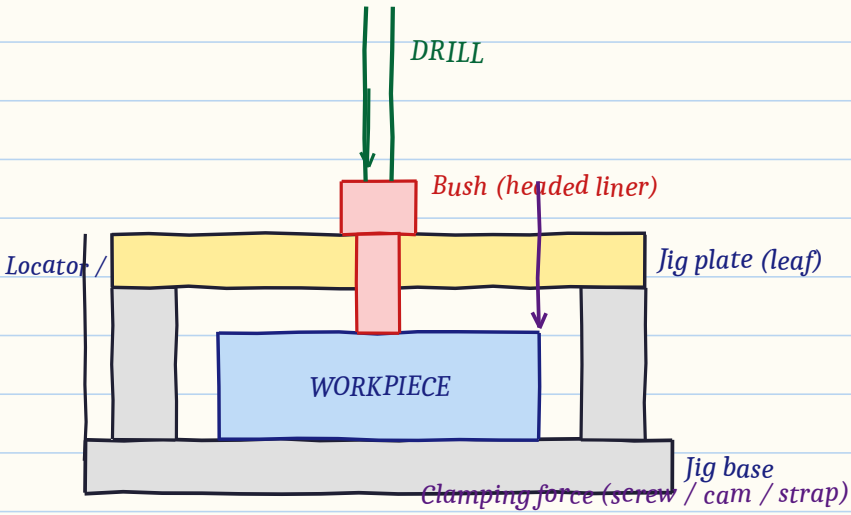
6.1 Types of clamps

Clamp	Working	Remarks
Strap (finger) clamp	bar pivoted over a heel; a nut/screw presses (lever of 1st, 2nd or 3rd class)	simple, cheap; for general use
Screw clamp	direct screw pressing the work; swivel pad	slow, high force
Cam clamp	eccentric or spiral cam on pin; rotating lever 90-180 deg	quick action ; self locking if angle < friction
Wedge clamp	taper wedge with 5 to 8 deg (self-locking)	quick; high force
Toggle (latch) clamp	links reach "dead-centre" so locked	very quick; low force; adjustable for thickness
Hook bolt / swing clamp	swings out of the way for loading	clear loading space
Equalising (floating) clamp	pivots to share force on 2-3 points	for uneven cast part
Hydraulic / pneumatic clamp	cylinder or diaphragm; pressure 4-7 bar	high production; automatic; CNC fixtures
Magnetic / vacuum chuck	holding by magnetism / suction	thin plates, grinding; no distortion

- Force relations: screw clamp: T = W x (r_m tan(alpha + phi) + mu_c x r_c), so W = T / (...) ; for a strap clamp with one end on work (L1) and bolt (L2) and heel (L3): clamp force on work = bolt force x L2/(L1 + L2) (bolt in the centre -> 50 %).
- Cam clamp: spiral curve rises uniformly; the rise angle must be less than the friction angle (about 5-8 deg) so that it self-locks; eccentric throw e = 1/2 (total rise) .
- Toggle: force at work = P / (2 tan(alpha)) - becomes very large near dead centre (alpha -> 0); used with an adjustable stop .

7. Drill Jigs

- A drill jig holds the part and guides the drill through a hardened bush to ensure position, perpendicularity and accuracy of holes, with no marking-out.



Parts of a drill jig

1. Jig body / plate
2. Locating elements
3. Clamping elements
4. Drill bushes (guide tool)
5. Feet (to rest on drill table)
6. Chip clearance & loading space

7.1 Types of drill jigs

Type	Features / use
Template jig	simplest: a plate with bushes laid on the work; for lightly built, large parts; low accuracy, low volume
Plate jig	plate with bushes + clamps and locators; accurate for flat parts; plate jig with feet clears chips
Open (table) jig	work is placed on a table; bush plate fixed; for parts drilled from one side
Leaf (hinged) jig	leaf hinged on a body, opens for loading; clamping by a thumb-screw/cam; small parts, drilling from one side

Type	Features / use
Box jig (tumble jig)	part enclosed, bush on several faces; box turned over to drill on different sides; heavy, rigid
Channel jig	U-shaped body, part fits into channel, clamped by screw or cam; simple; one or two sides
Angle plate jig	holes at right angle to the locating face; angle plate base
Indexing jig	drill several holes on a circle/straight line using index plate and plunger
Trunnion jig	large castings rotate on trunnions for drilling at several angles
Universal jig	adjustable/ modular for family of parts; with sub-plates
Pot jig / Diameter jig	cylindrical part held in a pot/V-block cradle; radial holes drilled
Pump jig	lever-operated (rack and pinion) quick lift of bush plate; for repetitive work

7.2 General considerations in the design of drill jigs

1. Study part drawing: material, size, quantity, operations, tolerance of holes and locating datum.
2. Provide clear access to load and unload the part with the leaf/ swing clamp; mistake-proof (one way only).
3. Clamp and locator must not interfere with tool path or the bush; the bush plate must be rigid ; thickness 1.5-2 x bush length to retain bushes (about 15-25 mm).
4. Hold the part against thrust of the drill - the clamp resists torque and the locator resists thrust : the cutting force goes into the base, not the clamp.
5. Provide chip clearance : a gap of 1 x drill dia (>= 1.5 mm for small) between bush end and workpiece; open at the bottom for chips; clearance slot / holes to remove chips; add coolant access.
6. Feet: 4 pad feet (or 3 for rock-free) at the base, ground flat; height >= drill extension, so that the drill's exit does not touch the table (also for through holes).
7. Keep jig light for handling (use cast aluminium for body) and easy to invert ; handles to lift; weight < 10 kg for hand use.
8. Rigid construction - fabricated welded, cast or built-up screwed and dowelled (two dowels each part).
9. Use standard components and hardened wear surfaces; provide identification marks ,

finishing, deburr.

10. Consider economics - cheapest design that meets accuracy; use jig only if number of parts justifies .
11. Fool-proof , safe, visible location of the work and inspection of locating surface cleanliness .

Tolerances in jigs

- Tolerance on tool dimensions = 20 - 50 % of the part tolerance (generally 1/5 to 1/3)
- Distance between bushes: part tolerance x 0.2 (+/- 0.02 for +/- 0.1 on the part)
- Bush axis perpendicularity: 0.01 / 25 mm

8. Drill Bushings

- Bushes guide the drill, fix its position and hold it against deflection. Made of hardened & ground tool steel (HSS / 60-64 HRC) or case-hardened steel; carbide for long run .

8.1 Types of drill bushes

Bush	Features
Press-fit bush (fixed): plain or headed	pressed in jig plate (H7/p6); for short run (not replaced); headed bush has flange to stop tool thrust
Renewable (replaceable) bush	used with a liner bush ; fixed by lock screw / clamp ; for long runs where wear needs replacement ; slip fit h6/G7
Slip (removable) bush	slip into liner; can be changed quickly for drill-ream-tap sequence on the same hole; with a lock screw or knurled/ latch
Screw bush	threaded into jig plate; adjustable for height; for clamping the part also
Special (large) bushes	Plate bush (rectangular), V-bush, angle bush , swivel / half bush for slanted surface , bush with extended nose for recess
Liner bush	a permanent bush press-fit in jig body to receive renewable bush

- Bush dimension (standard IS 3403): bore for drill: +0.010 to +0.025 (F7) above nominal size; bush length L = 1.5 to 2.5 x bush bore diameter (the longer, the better guidance); head dia = OD + 3 mm; the clearance between bush bottom and work = 0.3-1.0 x drill dia (for chip escape, but min. 1.5 mm).
- Bush bore: Ream bush (G6) , tap bush (H7) , drill bush (F7) : the bushes are given tolerances for the tool.

- Rule: material of the bush harder than tool shank ; surface finish Ra 0.4 in the bore; concentric to OD within 0.005 mm .

8.2 Methods of construction of jigs (jig bodies)

- Built-up (assembled) construction: parts of standard plate, bars, angles - screwed and dowelled together; cheap, quick; suits one-off.
- Welded (fabricated) construction: steel sections welded; light and quick; needs stress relieving & machining after.
- Cast construction: cast iron (grey) or cast steel / aluminium; good for complex shapes, damping & rigidity; pattern cost -> only for large quantity .
- Machined from solid / standardised modular systems: T-slot base plates and modular fixture kits (quick-change tooling) for flexibility.

Jig design (example check)

- Drill force (thrust) on a 10 mm drill in steel ~ 2000 N; torque ~ 12 N.m
- Locator and clamp must resist thrust (axial) and torque (rotation): clamp force x mu x radius >= T
- Add a safety factor of 2 to 3

9. Fixtures - Introduction and Types

- Fixtures are bolted to the machine table with tenon (key) blocks fitting in T-slots to align with the table; they are heavy & rigid to cope with large milling cutting forces and vibration .
- Setting: the cutter position is set with a set block (setting piece) and thickness (feeler) gauge of 2 to 3 mm to avoid cutting the set block (cutter never touches the fixture).
- Types of fixtures (by machine): milling, turning (lathe), boring, broaching, grinding, planing, shaping, welding, assembly, inspection fixtures; and by the power : manual, pneumatic, hydraulic.
- General design points: low centre of gravity , wide base, clamp over supports , chip clearance (large flushing holes), tenons , lifting lugs , wear plates, burr/cutter clearance .

9.1 Vice fixtures

- Use a standard machine vice with special soft jaws (shaped to the work) as the fixture : the vice acts as the clamp and base; cheap and quick.
- Types: plain vice, swivel vice, self-centring vice , hydraulic vice , tool-maker's vice , vice

with special inserts/V-jaws .

- Soft jaws (aluminium, mild steel, copper-alloy) machined to profile of part - jaw cavity cut in place for accuracy; replacable jaws .
- Design points: the part must rest on parallels or on a base; the movable jaw lifts slightly - use hold-down; the vice is aligned using a dial indicator or two tenons (keys) ; useful for small and medium parts, low cost .

9.2 Milling fixtures

- Used for slots, flats, key-ways, profiles, gear blanks . Must resist intermittent heavy cuts .
- Types: plain (single-part), string milling (a row of parts in a line, feed continuous), gang-milling (several cutters), straddle milling , indexing (rotary) fixtures, profile milling (cam/pattern controlled), rotary (continuous) fixtures (parts loaded on a rotating table or drum while cutting - reduces idle time to nearly zero).
- Design rules:

1. Clamp so that cutting force is directed on the fixed jaw/ locators (feed against the fixed locators ; cut in climb vs conventional).
2. Strong clamps - force at least 2x tool force; keep them close to the cut but out of the cutter path.
3. Rigid and heavy , low height, large base for stability; keep sections thick and avoid overhang .
4. Set blocks, tenons , ample chip clearance and flushing coolant , cutter overrun , and safe access for loading .
5. Fool-proof and lock-in so that the part cannot be loaded wrongly; wear plates and replaceable locators.

9.3 Boring fixtures

- Used on vertical / horizontal boring mills, lathes, jig-boring machines to bore holes of high accuracy (bearing bores, engine blocks, gear boxes).
- The tool (boring bar) is guided by bushes in the fixture (as in jigs) or the bar is supported at both ends in line with the spindle (to prevent deflection and chatter).
- Features: heavy base with tenons , locators ; clamps against cutting force ; bush (boring bar supports) with oil holes ; piloted boring bar with floating holder ; provisions for chips .
- Lathe boring fixtures: the fixture is mounted on the face plate (angle plate type) with counterweight to balance the rotating mass -> safety : no projecting parts, guards.

9.4 Broaching fixtures

- A broaching fixture holds the workpiece against a large pull / push force while the broach moves through - usually a simple plate (seating plate with a hole larger than the broach teeth) fixed on the broaching machine face .
- Types: (1) Fixed plate (face plate) with hole/ spherical seat : the part rests against the plate; the spherical seat floating (self-aligning) lets the part to align itself with the broach (e.g. for keyway or hole broaching). (2) Surface broaching fixtures with clamps , holding the part in a flat-bed and the cutting thrust is taken by the rigid body . (3) Rotary / indexing broaching fixtures for several surfaces (e.g. splined holes).
- Design points: the thrust (up to many tonnes) is taken on a strong seat ; clamp needed only for surface broaching; chips fall freely (slotted opening); guide the broach into the hole (pilot bush); load by hand easily ; fixtures should withstand shock and position stably .

Jigs vs fixtures - exam tips

- Jig -> bushes, light, drilling; Fixture -> set block, heavy, milling
- Locate by 3-2-1; clamp over locators; keep chip clearance
- Always mention fool-proof , standard parts , economy

10. Quick Revision - Unit III

- Taylor's principle: GO full form, NO-GO one dimension at a time. Gauge tolerance = 10 % of work tolerance.
- Indicating gauges: dial, mechanical, optical, electrical, pneumatic. Automatic gauges: in-process / post-process.
- 3-2-1 principle: 3 + 2 + 1 points; locating elements: pins, buttons, V-blocks, nests, mandrels.
- Clamps: strap, screw, cam, wedge, toggle, hydraulic; clamp force towards the locators.
- Drill jigs: template, plate, leaf, box, channel, indexing. Bushes: fixed, renewable, slip, liner; L = 1.5 - 2.5 d.
- Fixtures: vice, milling, boring, broaching - bolted to table; set block + feeler gauge.